

useful features

useless features

**Linear
regression**

$\beta_1 \cdot \text{area}$

$\beta_2 \cdot \text{rooms}$

$\beta_3 \cdot \text{door no.}$

$\beta_4 \cdot \text{birth month}$

every feature kept,
full-size coefficients

Ridge

$\beta_1 \cdot \text{area}$

$\beta_2 \cdot \text{rooms}$

$\beta_3 \cdot \text{door no.}$

$\beta_4 \cdot \text{birth month}$

every feature kept,
all coefficients shrunk

Lasso

$\beta_1 \cdot \text{area}$

$\beta_2 \cdot \text{rooms}$

~~$\beta_3 = 0$~~

~~$\beta_4 = 0$~~

useless features removed
(coefficient exactly 0)